Numerical Example:— ①

x	1.00	2.00	3.00	4.00	5.00	→ Input data (Independent variable)
y	1.00	2.00	1.30	3.75	2.25	→ Target data (dependent variable)

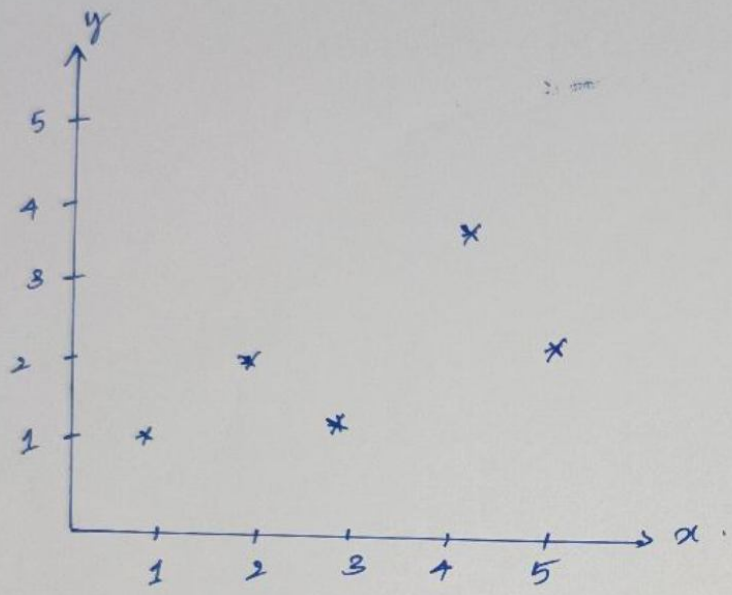

Figure 1: Scatter plot for the given data.

⇒ Aim is to find the regression line for the given data

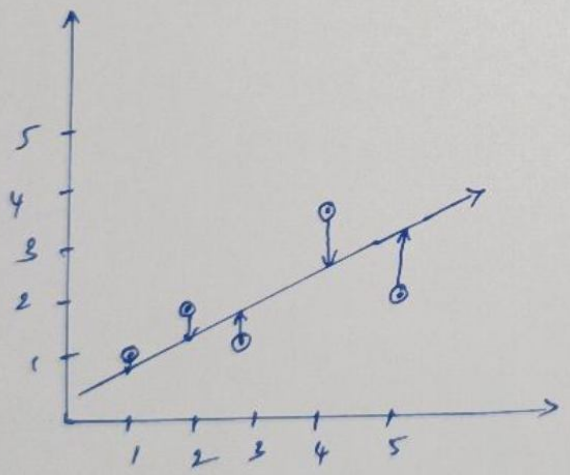

Figure 2: Reg. line

⇒ Straight line can be represented by

$$\hat{y} = bx + A$$

$b \rightarrow$ slope of the reg. line

$A \rightarrow$ Intercept value with y-axis.

$x \rightarrow$ Input

$\hat{y} \rightarrow$ prediction Score.

⇒ To find the reg. line we need to find the values of 'b' and 'A'.

where, $b = r\, S_y / S_x$ (or) $r \dfrac{\sigma_y}{\sigma_x}$

and

$$A = \mu_y - b \cdot \mu_x$$

Here, S_y - standard deviation of 'y' (or σ_y)

S_x - standard deviation of 'x' (or σ_x)

μ_y - mean of 'y'

μ_x - mean of 'x'

r - correlation.

~~and~~ To find b and A we need to find S_y, S_x, μ_y, μ_x and r

x	y	$X = x - \mu_x$	$Y = y - \mu_y$	XY	$X^2 = (x-\mu_x)^2$	$Y^2 = (y-\mu_y)^2$
1.00	1.00	−2	−1.06	2.12	4	1.1236
2.00	2.00	−1	−0.06	0.06	1	0.0036
3.00	1.30	0	−0.76	0	0	0.5776
4.00	3.75	1	1.69	1.69	1	2.8561
5.00	2.25	2	0.19	0.38	4	0.0361
$\mu_x = \frac{\Sigma x_i}{n} = 15/5 = 3$	$\mu_y = \frac{\Sigma y_i}{n} = 2.06$			$\Sigma XY_i = 4.25$	$\Sigma (x_i - \mu_x)^2 = 10$	$\Sigma (y_i - \mu_y)^2 = 4.597$

$\mu_x = 3$; $\mu_y = 2.06$.

<u>Formula for S.D:—</u>

$$S.D. \sqrt{\sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N-1}}$$

$$S_x = \sqrt{\sum_{i=1}^{5} \frac{(x_i - \mu_x)^2}{5-1}} = \sqrt{\frac{10}{4}} = \sqrt{2.5} = 1.5811$$

$$S_y = \sqrt{\frac{\sum_{i=1}^{5} (y_i - \mu_y)^2}{5-1}} = \sqrt{\frac{4.597}{4}} = 1.072.$$

Formula for Correlation :-

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \; \Sigma y^2}}$$

$$r = \frac{4.25}{\sqrt{10 \times 4.597}} = \frac{4.25}{6.7801}$$

$$r = 0.6268 \simeq 0.627$$

Subtitute the values of μ_x, μ_y, S_x, S_y, and r in the formula's of 'b' and 'A'

$$b = r \cdot S_y / S_x = (0.627) \times (1.072) / 1.581$$

$$= 0.672 / 1.581$$

$$b = 0.425$$

$$A = \mu_y - b \cdot \mu_x = (1.072) - (0.425)(3)$$

$$= 2.06 - 1.275$$

$$A = 0.785$$

Now sub. the value of 'b' and 'A' in $\hat{y} = bx + A$

$$\therefore \quad \hat{y} = (0.425)\,x + 0.785 \Rightarrow \text{Required regression line.}$$

For $x = 1$, $\hat{y} = 0.425(1) + 0.785$

$$= 1.210$$

For $x = 2$, $\hat{y} = 0.425(2) + 0.785$

$= 1.64$

For $x = 3$, $\hat{y} = 0.425(3) + 0.785$

$= 2.060$

For $x = 4$, $\hat{y} = 0.425(4) + 0.785$

$= 2.485$

For $x = 5$, $\hat{y} = 0.425(5) + 0.785$

$= 2.91.$

Then the calculation of the error term:-

x	y	$\hat{y}$	$y - \hat{y}$	$(y - \hat{y})^2$
1.00	1.00	1.210	-0.210	0.044
2.00	2.00	1.635	0.365	0.133
3.00	1.30	2.060	-0.760	0.578
4.00	3.75	2.485	1.265	1.600
5.00	2.25	2.91	-0.66	0.436

The regression line $\hat{y} = 0.425x + 0.785$

For each input value of 'x', respective $\hat{y}$ values are predicted.